Muyang Chen

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EDUCATION

Peking UNIVERSITY

Bachelor of Physics

- **Academics:** overall GPA: 3.81/4.0
- **Graduate Courses:**
  Quantum Many-Body Theory, Quantum Statistical Physics, Group Theory I, Group Theory II, Quantum Field Theory, General Relativity, Solid State Theory, Advanced Quantum Mechanics

Honors and Awards

- Merit Student(2022,2024)
- Wang Chenyang-Cheng Ya Educational Fund(2024) (9 awardees per year for top12 students)
- Fourth Place in the National Collegiate Floorball Championship(2024)
- Beijing Municipal Natural Science Foundation (2023)
  (Research projects of top14 undergraduate students will be funded by this foundation.)
- Scholarship for WeiMing Physics Students(2021,2022,2023)
- The Third Prize Scholarship(2023)
- SK Scholarship(2022) (1 awardee per year for top14 students)
- Third Prize in CUPT(China Undergraduate Physics Tournament) (2022)
- Freshman Scholarship(2021) (20 awardees per year)
- Chinese Physics Olympiad Gold Medal(2021)
  (Chinese National Training Team)

FIELDS OF INTEREST

- Strong Correlation Physics in Twisted Materials
- Classification and Response Theory of Topological States
- Transport Problems in Disorder Systems

RESEARCH EXPERIENCE

Hartree-Fock Study of Twisted Bilayer WSe2

Advisor: B.A. Bernevig, Princeton University 2024.4– Present

- Performing self-consistent Hartree-Fock calculations for twisted bilayer WSe_2
- Exploring the effects of varying displacement field and interaction to uncover the mechanisms underlying potential superconductivity
- Investigating transitions between correlated insulator and metal observed in the experiment, aiming to provide theoretical predictions and insights

Classification of Magnetic Topological States with C_4T symmetry

Advisor: Zhida Song, Peking University 2023.6– 2024.1

- Developed methodologies to derive topological invariants for characterizing states that are indistinguishable via symmetry indicators or topological quantum chemistry
- Introduced a Wilson loop-based approach to identify fragile topological states protected by specific crystalline symmetry
- Utilized symmetry eigenvalues and modified Wilson loop to distinguish obstructed atomic insulating phases from atomic insulating phases

Quantum Geometry in Topological Band Theory

Advisor: Zhida Song, Peking University 2023.4 – 2023.6

- Utilized the semi-positive definite property of quantum geometric tensor to detect the lower bound of Wannier spread in Hopf Insulator
- Explored the relationship between Hopf invariants and their corresponding Wannier spread, providing new perspectives on quantum geometric properties in topological materials

SKILLS

- **Programming:** MATLAB, Mathematica, Python, LaTeX
- **Language Proficiency:** GRE general: 332, TOEFL iBT: 103